

# ECE 0402: Signals, Systems & Probability

## Group Project Guidelines

### Objective

The goal of this project is to deepen your understanding of signals and systems by applying core concepts to real-world problems, simulations, or case studies.

### Project Requirements

Your project/presentation must follow these rules:

- Send me your slides **at least three days** before the presentation day
- Focus on a well-defined problem or application
- Use at least one core concept covered in the lecture, such as:
  - Convolution
  - Fourier Transform
  - Sampling and Aliasing
  - LTI Systems
  - Filtering
- Include at least one demo that can be understood by the class in 2 minutes, such as:
  - Noisy vs. filtered audio
  - Aliasing visualization
  - Simulation of a system
- **Avoid copying AI-generated content or textbook derivations without explanation**
- Divide responsibilities clearly and explicitly state who did what
- Provide references (indicate where you obtained the information)

## Suggested Topic Areas

You may choose one of the following topics or propose your own:

- Audio and Speech Signals
  - Designing a Noise Reduction Filter for Voice Signals
  - How Auto-Tune Works: Pitch Detection and Modification
  - Echo and Reverberation: Modeling with Convolution
  - Speech vs Music: Signal Characteristics in Time and Frequency
- Image and 2D Signals
  - Image Blurring and Sharpening Using 2D Convolution
  - Edge Detection as a Filtering Problem
  - Image Compression Basics (Why JPEG Works)
- Sampling and Digital Signals
  - Aliasing in Digital Audio: When Sampling Goes Wrong
  - Anti-Aliasing Filters in Practice (Before Sampling)
  - Reconstruction of Signals from Samples (Interpolation Methods)
- Communication Systems
  - Amplitude Modulation (AM): How Radio Signals Are Transmitted
  - Frequency Modulation (FM) and Noise Robustness
  - How Noise Affects Communication Systems
- Systems and LTI Insight
  - Understanding Convolution Through Real Systems (Audio Reverb)
  - System Identification: Estimating an Unknown System
  - Causality and Stability in Real Filters
- Real-World Signals and Systems
  - ECG Signal Analysis: Detecting Heartbeat Patterns
  - Noise Removal in Biomedical Signals
  - How Noise-Canceling Headphones Work
  - Signal Processing in Radar or Sonar Systems
  - How Shazam Identifies Music (Basic Signal Matching)

## Resources (should be properly cited)

- Lecture slides and textbook
- *Signals and Systems Using MATLAB*, 4th edition, by Luis F. Chaparro
- *Linear Systems and Signals*, 2nd edition, by B. P. Lathi
- 3Blue1Brown (available on Bilibili)
- MIT OpenCourseWare
- AI tools (e.g., ChatGPT, DeepSeek)

## Grading Rubric

- Meeting all project requirements: 40%
- Understanding of concepts: 30%
- Clarity of presentation: 20%
- Q&A performance: 10%

## Final Advice

- Avoid overly broad topics like “Fourier Transform” or “Sampling”. Instead, choose focused topics such as “Aliasing in Audio Signals and How to Prevent It.”
- Feel free to explore different/creative ways to present your topic. Think from the audience’s perspective: What will help them understand the most?
- A good presentation takes planning and practice. Make a schedule and start early.
- Be sure to **test the device before your presentation day** to avoid any technical issues.